

Two-Way Fixed Effects and Modern Difference-in-Differences: An Empirical Comparison

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This note compares two-way fixed effects (TWFE) estimates from top general-interest journals in economics with estimates from the group-time difference-in-differences estimator of Callaway and Sant’Anna (2021; CSA) and the imputation estimator of Borusyak, Jaravel, and Spiess (2024; BJS). We use recovered observation-level data and focus on binary treatments with staggered adoption. Sign reversals are relatively rare, and especially so with statistically significant differences.

Sample and implementation. We (the author and ChatGPT Astra) screened 5,014 bibliographic records from 2016–2025 in five economics journals: AER, Econometrica, JPE, QJE, and ReStud. Screening was for published TWFE or CSA/BJS estimates with a single binary treatment and staggered adoption; this advanced 53 papers to package and data review; four further papers remained held for author verification. We collected qualifying main-text results from tables and figures, allowing panels, repeated cross sections, and fixed baseline characteristics interacted with time effects; other time-varying controls were excluded. The final sample includes 351 specifications from 24 papers with available replication data.

Each scalar regression contributes one difference; each event study contributes a post-treatment average over commonly estimable published horizons or bins and an analogous pre-treatment average when available. Baseline TWFE retains the published sample, including always-treated observations, subject to documented exceptions. Modern estimators use their own identifiable treated samples. We also considered an “aligned” comparison which refits TWFE on the exact union of observations used by the corresponding modern estimator. We preserve the published clustering conventions and estimate standard errors and cross-estimator covariances using a joint weighted bootstrap. Appendices A-E detail the sample construction and analysis.

Some specifications were excluded due to small effective sample sizes (i.e., high leverage). This leaves 285 distinct specifications from 22 studies: 257 scalar comparisons and 28 post-treatment event-study averages. The event studies also contribute 22 separate pre-treatment averages, yielding 307 summaries. These include 273 CSA comparisons and 297 BJS comparisons.

Results. Figures 1 and 2 plot each modern estimate against TWFE. Both axes scale by the TWFE standard error; points outside $[-5, 5]$ are capped. Filled markers identify a pointwise significant paired difference at the 5% level; crosses indicate high-leverage estimates. Circles denote static specifications; squares denote event-study averages.

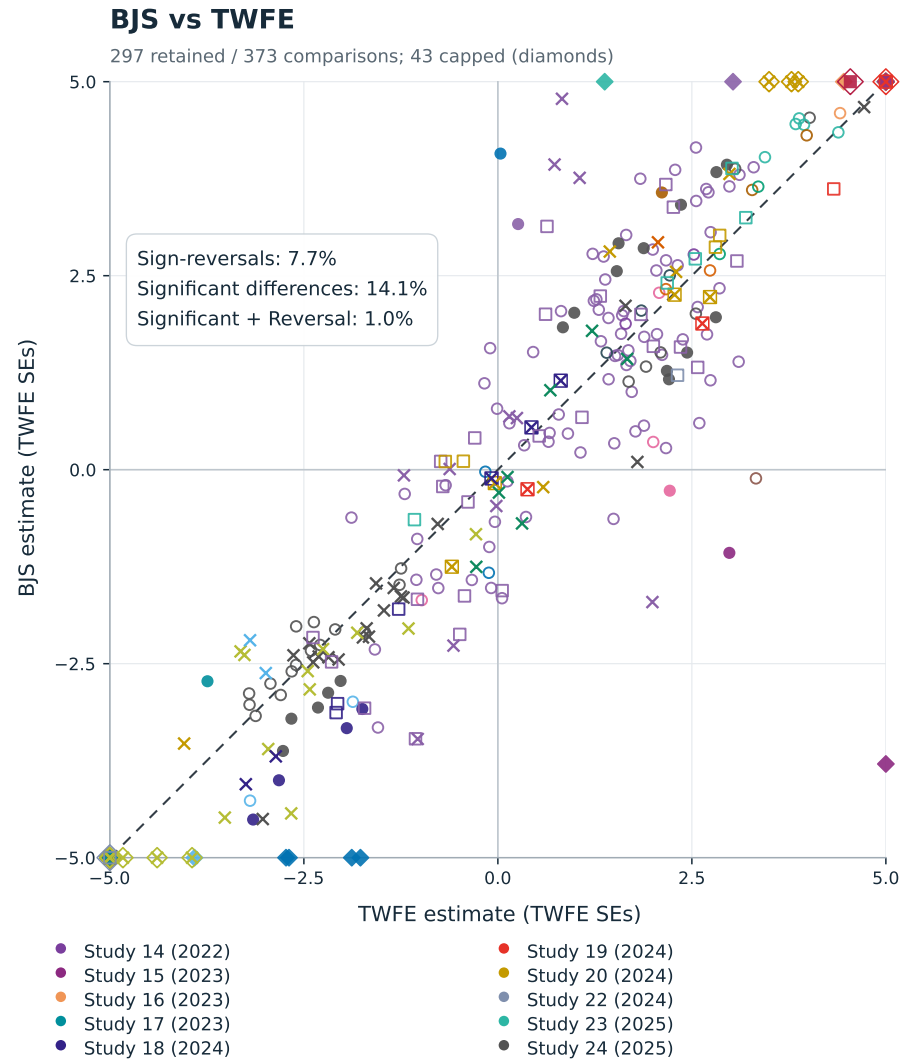
In the baseline sample, opposite signs occur in 13.6% of CSA pairs and 7.7% of BJS pairs; significant differences occur in 8.4% and 14.1%, respectively. Both conditions hold in 1.5% and 1.0%. With aligned samples, the corresponding shares are 16.5% and 6.4% for opposite signs, 7.7% and 10.1% for significant differences, and 0.0% and 1.0% for both. These shares weight retained pairs equally, including pre-treatment averages and warning-marked pairs in the denominator.

On average across retained comparisons, CSA and BJS standard errors are 49% and 26% larger than their corresponding TWFE standard errors in the baseline analysis; for the aligned sample they are 28% and 13% larger.

¹Contact: peter_hull@brown.edu. I thank Jiafeng Chen and Jonathan Roth for helpful comments. OpenAI’s ChatGPT did all of the heavy lifting. All errors in prompting or review are my own. Replication data and code is available at <https://github.com/peterdhull/twfe-replication>.

Figure 1: Baseline comparison: eligible original samples

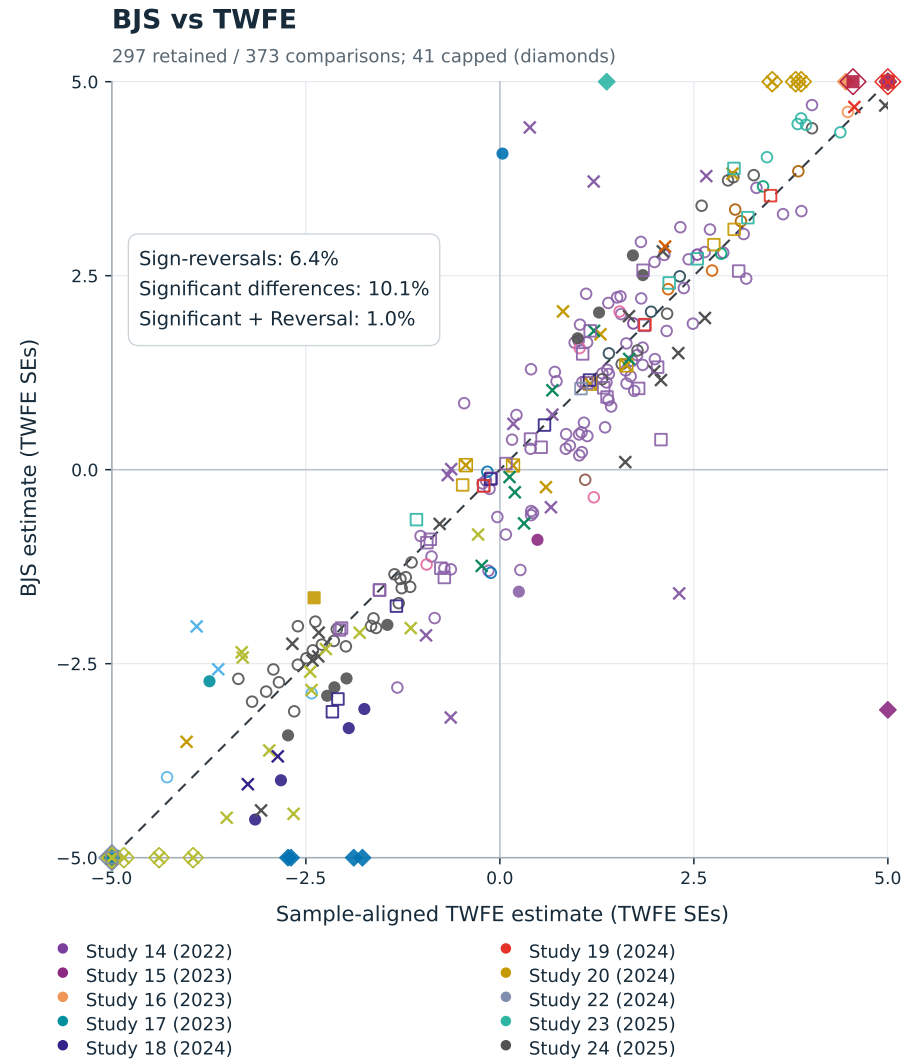
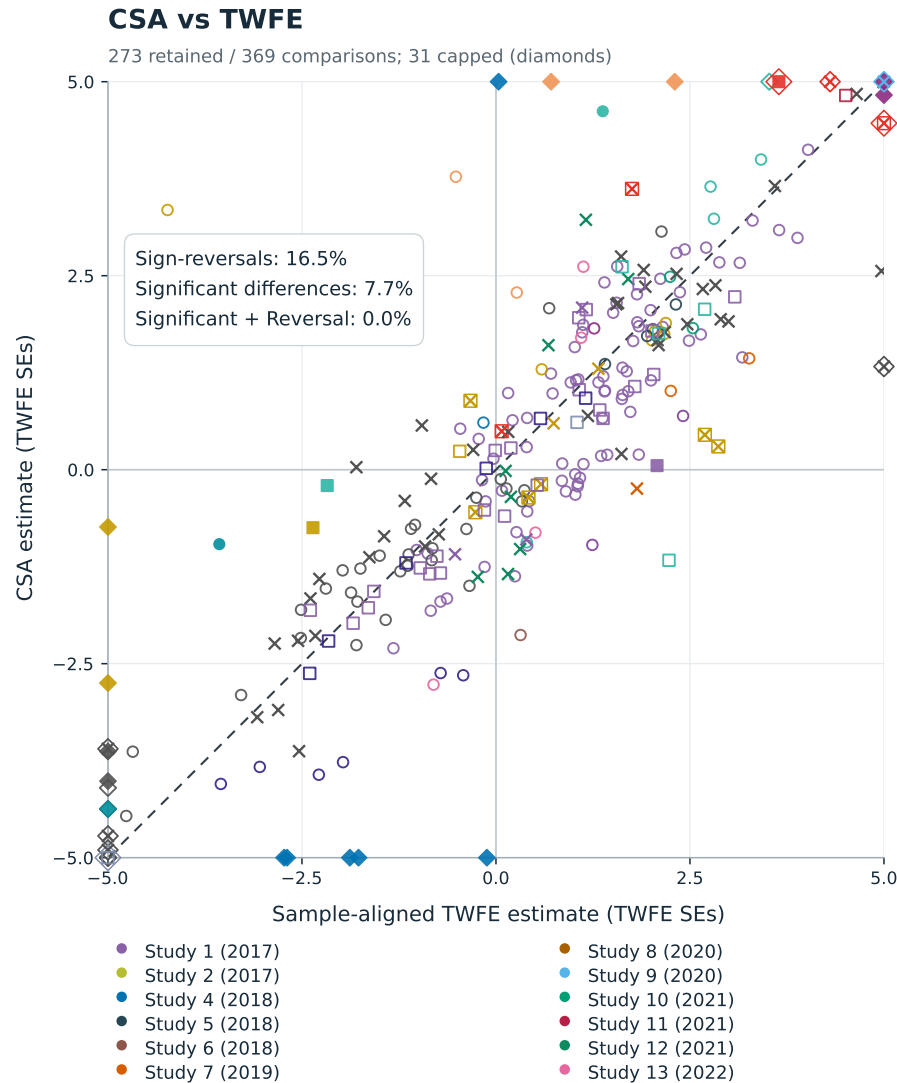
22 studies · Filled: paired $|t| > 1.96$ · Hollow: otherwise · Cross: fragile inference



Notes: Circles: static specifications. Squares: event-study averages, with pre and post entering separately. Pre averages are diagnostics, not treatment effects. Shares use retained comparisons, including flagged points. Significant + Reversal: opposite nonzero signs and a filled marker. Diamonds: coordinates capped at ± 5 ; displayed gaps may differ from actual gaps. Excluded if either estimate has effective clusters < 5 or maximum cluster leverage $> 50\%$. Crosses: effective clusters < 10 or maximum leverage $> 25\%$ for either estimate or their difference, or prior inference warnings; significance fill suppressed. Leverage uses squared outcome-linear weights by the existing bootstrap clusters; it is a design diagnostic, not a variance guarantee. TWFE retains its eligible sample; modern estimators use their own supported targets. Both axes divide by this pair's saved TWFE SE; the plotted gap is not the paired-difference t statistic. Common horizons are unchanged. Fill uses paired-bootstrap normal tests, without multiplicity adjustment. Dashed line: equality. Study 8 uses unrestricted TWFE; its restricted published fit is retained separately. Study 4 clusters 143 builders despite only three treatment-rollout regions.

Figure 2: TWFE refitted on estimator-specific aligned samples

22 studies · Filled: paired $|t| > 1.96$ · Hollow: otherwise · Cross: fragile inference



Notes: Circles: static specifications. Squares: event-study averages, with pre and post entering separately. Pre averages are diagnostics, not treatment effects. Shares use retained comparisons, including flagged points. Significant + Reversal: opposite nonzero signs and a filled marker. Diamonds: coordinates capped at ± 5 ; displayed gaps may differ from actual gaps. Excluded if either estimate has effective clusters < 5 or maximum cluster leverage $> 50\%$. Crosses: effective clusters < 10 or maximum leverage $> 25\%$ for either estimate or their difference, or prior inference warnings; significance fill suppressed. Leverage uses squared outcome-linear weights by the existing bootstrap clusters; it is a design diagnostic, not a variance guarantee. Exact-union sample alignment does not equate estimator weights or estimands. Both axes divide by this pair's saved TWFE SE; the plotted gap is not the paired-difference t statistic. Common horizons are unchanged. Fill uses paired-bootstrap normal tests, without multiplicity adjustment. Dashed line: equality. Study 8 uses unrestricted TWFE; its restricted published fit is retained separately. Study 4 clusters 143 builders despite only three treatment-rollout regions.

A Search frame, eligibility, and the implemented sample

A.1 Bibliographic frame and screening

The frozen search frame was assembled on 1 October 2026 from Crossref journal-article records for the *American Economic Review* (AER), *Econometrica*, *Journal of Political Economy* (JPE), *Quarterly Journal of Economics* (QJE), and *Review of Economic Studies* (ReStud). Of 5,121 retrieved records, 5,014 fall within the configured 2016–2025 publication window; 107 date spillovers or unresolved-date records were retained outside that denominator. The dating rule prioritizes the print publication date, followed by published, issued, and online dates. Missing month or day components are assigned January or the first day for filtering. The retrieval contained no duplicate losses under the DOI/paper-identity rule.

Records were reviewed in a frozen hash-based order. All 5,014 positions have a recorded disposition, but screening depth varies: a first-pass exclusion is not a full-text replication audit. Text searches and external catalogs supplied leads, followed by inspection of equations, main-text tables and figures, and replication materials when necessary. The frame has not been independently reconciled against every journal’s table of contents; complete coverage of this frozen frame therefore does not establish exhaustive coverage of all eligible published work.

Of the 53 advancing papers, 50 entered through a qualifying TWFE result and three through a modern-estimator-only result. The screening inventory at that checkpoint contains 1,117 candidate scalar specifications or event-study vectors across the 57 advancing or held papers: 842 TWFE scalars, 216 TWFE vectors, 11 modern scalars, and 48 modern vectors. Of these candidates, 1,067 belong to advancing papers and 50 to hold-only papers; 248 candidate rows were held, including 198 within advancing papers.

Table 1: Screening flow and current analysis coverage

Stage	Count
Retrieved bibliographic records	5,121
Out-of-window or unresolved-date records	107
In-window records with screening dispositions	5,014
First-pass exclusions	3,419
Equation-level exclusions	1,531
Exclusions following full-text audit	7
Papers held for verification	4
Papers advanced to package/data review	53
Implemented studies / distinct specifications	24 / 351
Studies / specifications retained in at least one panel	22 / 285
Additional retained pre-treatment averages	22
Retained CSA / BJS estimator pairs	273 / 297

Screening dispositions are the validated 3 October checkpoint. Implementation and figure counts refer to the 9 October release. Later stages are not mutually exclusive exclusion categories: a paper can advance while individual specifications remain unresolved.

A.2 Journal and publication-year coverage

The implemented source release contains 351 distinct specifications from 24 studies; 285 specifications from 22 studies retain at least one plotted modern-estimator comparison. Table 2 summarizes this coverage without identifying studies. The source specifications comprise 323 scalars and 28 post-treatment event-study averages; adding the 22 available pre-treatment averages yields 373 source summaries and 307 retained summaries. The observed publication range is 2017–2025 within the 2016–2025 screening window.

Table 2: Implemented and retained coverage by journal and publication year

	Implemented		Retained	
	Studies	Specifications	Studies	Specifications
<i>Journal</i>				
AER	10	148	9	110
Econometrica	3	28	3	28
JPE	7	155	6	133
QJE	3	18	3	12
ReStud	1	2	1	2
Total	24	351	22	285
<i>Publication year</i>				
2016	0	0	0	0
2017	3	140	2	118
2018	3	11	3	11
2019	1	3	1	3
2020	2	11	2	11
2021	3	17	3	11
2022	2	6	2	6
2023	3	9	3	9
2024	5	53	4	27
2025	2	101	2	89
Total	24	351	22	285

Retained coverage means that at least one estimator pair survives the concentration screen. Distinct specifications are counted once regardless of the number of estimator pairs or available pre-treatment averages. These implementation counts do not revise the frozen screening checkpoint.

A.3 Specification eligibility and amendments

Eligible specifications have one binary treatment at a stable assignment-unit level, unit or group effects, and calendar-time effects. At least two distinct first-adoption dates must be observed among treated units in the usable sample. Panels and repeated cross sections are both eligible.

The adjustment rule permits fixed pretreatment characteristics, including flexible functions of those characteristics, interacted with calendar-time effects, as well as fixed subgroup-by-time effects. Ordinary time-varying controls, changing subgroup membership, unit-specific trends, additional

treatments, treatment-heterogeneity interactions, triple differences, instrumental variables, nonlinear outcome models, and stacked event-study specifications are outside the screen.

The target inventory includes all qualifying main-text outcomes, transformations, samples, weighting choices, and specifications in both tables and figures. Appendix-only results do not provide an entry. A duplicate regression appearing in a table and a figure is one specification with multiple source locations. An inference-only variant is not a new point-estimator specification. A dynamic regression is inventoried as one coefficient vector. A static effect displayed beside that same vector is a linked estimand unless it uses a distinct sample or weighting definition. A small number of published TWFE specifications include treatment reversals; we drop these observations.

B Estimation samples, estimators, and event-study summaries

B.1 Baseline TWFE and outcome normalization

For a static specification, the reference model has the form

$$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + q'_{it}\gamma + u_{it}, \quad (1)$$

where q_{it} contains only the permitted baseline-by-time or fixed-subgroup-by-time adjustments. The original outcome transformation, analysis weights, fixed-effect identifiers, and eligible sample are preserved. In repeated cross sections, i denotes the stable group used by the source specification where appropriate. Conflicting source records are not automatically collapsed or recoded; treatment-history identifiers and fixed-effect identifiers can differ when the source design requires it.

The baseline comparison TWFE imposes no CSA/BJs support restriction. It retains always-treated observations and observations lacking a measured untreated outcome. Approved post-adoption zero-treatment cells are omitted as above. Consequently, some baseline TWFE estimates differ from the published estimate. Event averages, modern-only entry, and the fixed-effect exception in Appendix E further cause these to differ.

Let $\mathcal{O}$ be the complete original author-estimation sample for a specification. The outcome scale is

$$s_Y^2 = \frac{\sum_{i \in \mathcal{O}} (Y_i - \bar{Y}_{\mathcal{O}})^2}{|\mathcal{O}| - 1}. \quad (2)$$

All comparison estimates use Y_i/s_Y , without subtracting the mean. Regression analysis weights do not enter this standard deviation. The original sample is used before reversal or estimator-support omissions, and aligned estimators retain this same scale. Previously logged or otherwise transformed outcomes remain in their published definition.

B.2 CSA comparisons

The core implementation constructs cohort-time effects using weighted cross-sectional means. For treatment cohort g , current period $t \geq g$, and an allowed baseline stratum h , let $b(g)$ be the latest observed calendar period before adoption and let C_t denote never-treated or not-yet-treated observations, with first adoption strictly after t . Without continuous adjustment, the comparison is

$$\widehat{ATT}_{hgt} = (\bar{Y}_{hgt} - \bar{Y}_{hgb(g)}) - (\bar{Y}_{hC_t t} - \bar{Y}_{hC_t b(g)}). \quad (3)$$

Each of the four means must have positive analysis-weight support. A supported stratum is retained even if another stratum for that cohort and time is unsupported. Effects are aggregated using the current treated-observation analysis-weight mass of supported cells.

This is a repeated-cross-section implementation even when the source data are a panel: it does not require every treated observation to have its own observed baseline outcome. Cohort-stratum baseline support can suffice. Its interpretation therefore requires the relevant composition and parallel-trends conditions; it is not universally the same statistic as a native balanced-panel CSA estimator. For allowed continuous baseline covariates interacted with time, the conditional branch uses outcome-regression DiD of the repeated-cross-section form, evaluating untreated change at pooled pre/post treated covariate means.

B.3 BJS comparisons

BJS fits the allowed untreated-outcome model using untreated observations only, imputes the untreated outcome at identifiable treated observations, and averages observed-minus-imputed outcomes using treated analysis weights. In the basic fixed-effect case, identification requires the relevant unit and time components to be connected through the untreated-observation graph. With additional permitted adjustments, prediction estimability is checked in the fitted design span.

BJS does not inherit CSA’s four-cell requirement, and its targets are not intersected with CSA’s. Conversely, CSA’s cohort baseline support does not guarantee an identifiable BJS unit-specific untreated prediction. The supported treated populations can therefore differ for reasons beyond the removal of always-treated observations. The row masks, support counts, and retained treated-weight shares are recorded at the specification-estimator level.

B.4 Exact-union sample alignment

For each estimator and contrast, the aligned comparison refits the full registered TWFE specification on the exact union of rows actually used by that modern estimator. For CSA, this includes treated-current, treated-baseline, donor-current, and donor-baseline rows for the selected effects and reference components. For BJS, it includes supported treated target rows and the eligible untreated rows used to fit its nuisance model.

The modern estimator is retained on its own target and construction; TWFE keeps the original specification, permitted adjustment span, analysis weights, outcome scale, and dependence rule when refitted on the union. In particular, alignment does not force a common distribution of implicit treatment-effect weights.

B.5 Event studies and reference bins

The complete joint event-study regression is retained. Its post-treatment summary is a linear contrast $\hat{\theta} = \sum_{k \in K} v_k \hat{\beta}_k$, with equal weights across registered published coefficients or bins unless different weights are explicitly recorded. An open-ended bin receives one bin’s weight; it is not split into individual years. Reported horizon coefficients do not count as separate specifications.

The comparison window is the structurally estimable intersection of registered horizons across the estimators included in that comparison. Default comparisons use TWFE, CSA, and BJS; a genuinely

single-modern comparison uses that estimator and TWFE and is labeled accordingly. Registered weights are renormalized over retained horizons. Availability is determined before bootstrapping from support and contrast estimability, not effect size, significance, estimated standard errors, or draw success. Individual omitted horizons and their reasons are preserved.

For aligned event studies, the baseline common window is the starting point. Exact-union masks are rebuilt and horizons unidentifiable in either corresponding TWFE fit are removed until the window stabilizes. Unpooled nuisance event indicators remain in the joint regression. One adjusted specification in Study 19 averages years 0–6 in the baseline family and 0–2 in the aligned family; its original registered range was 0–8. All other currently retained comparisons keep the same horizon set across families. Thus that point’s movement reflects both row alignment and a changed horizon.

Published reference bins are preserved. When the reference includes treated periods, modern effects are contrasted against the corresponding estimated reference quantity rather than relabeled as an untreated-baseline ATT. Its uncertainty and covariance enter every draw. This applies to Study 20’s omitted interval $(-10, 0]$, which includes adoption year zero.

Pre-treatment coefficients are pooled separately over common, strictly pre-treatment terms, excluding the omitted term and any bin containing treatment or post-treatment observations. In the core event implementation, CSA uses its pre-adoption baseline and donors untreated at both comparison dates; BJS pre coefficients come from a separate untreated-only joint event regression. The mixed-reference construction described above is a distinct contrast. These averages are retained as diagnostics and enter Figures 1–2 as separate points from the post-treatment averages. Both pre and post averages use square markers, while static specifications use circles.

Comparable pre-treatment averages are available for 22 of the 28 event-study specifications, across five studies. The remaining six specifications have no comparable modern pre-treatment estimate or no separately registered pre-treatment coefficient. Pre-treatment comparisons use their own pooled-contrast leverage measures and paired bootstrap covariance, with the same screening and inference rules as the other points. Missing pre-treatment estimates are not assigned zero. For ordinary untreated-only lead regressions, aligned TWFE and BJS may be algebraically identical. A pooled pre average is not an omnibus pre-trend test, and a mixed-reference pre average is not a pure untreated-baseline placebo.

C Joint inference and figure construction

C.1 Positive-weight cluster bootstrap

For each declared dependence block containing G clusters, draw independent $E_c^{(b)} \sim \text{Exp}(1)$ and set

$$W_c^{(b)} = \frac{GE_c^{(b)}}{\sum_{d=1}^G E_d^{(b)}}, \quad b = 1, \dots, 999. \quad (4)$$

The same cluster multiplier weights every observation and compatible estimator/specification coordinate in the block. Original analysis weights are multiplied by $W_c^{(b)}$; cluster totals are not equalized. Estimator fits, nuisance fits, aggregation masses, and the original-sample outcome scale are recalculated in each draw. Shared underlying innovations and recorded normalizations couple subsamples when their native cluster frames differ. Published dependence definitions, including observation-level robust inference where applicable, are preserved.

For the outcome scale alone, the draw-specific frequency-weight continuation is

$$s_Y^{2(b)} = \frac{\sum_{i \in \mathcal{O}} W_{c(i)}^{(b)} (Y_i - \bar{Y}_W^{(b)})^2}{\sum_{i \in \mathcal{O}} W_{c(i)}^{(b)} - 1}, \quad \bar{Y}_W^{(b)} = \frac{\sum_{i \in \mathcal{O}} W_{c(i)}^{(b)} Y_i}{\sum_{i \in \mathcal{O}} W_{c(i)}^{(b)}}. \quad (5)$$

Regression analysis weights are absent from this scale calculation. This convention returns the ordinary sample variance at unit multipliers. For event summaries, the fixed contrast, including any reference-bin subtraction, is applied in each draw. Marginal coefficient standard errors are never averaged to obtain an event-average standard error.

Strictly positive multipliers preserve the observed support instead of deleting clusters. No trimming, weight floors, ridge penalties, redraws, or failed-draw deletion are used. A coordinate's standard error requires all 999 finite draws; a covariance requires completeness for both coordinates. Point estimates are the unit-multiplier fits, not bootstrap means. Covariances are sample covariances centered on draw means with divisor 998 and no additional finite-cluster multiplier.

The source release has 25 declared covariance blocks across 24 studies, containing 2,957 coordinates with all 999 draws. These coordinates include auxiliary native estimates, event coefficients, pooled-pre contrasts, and archived alignment coordinates; they are not a count of active plotted summaries. Cross-specification covariance is available within a shared block. Study 2 uses two distinct published clustering partitions and retains two separate blocks; their cross-block covariance is unavailable, not zero. Between-study covariance is not estimated. Equal numerical draw indices from different blocks must not be treated as paired draws. These distinctions do not prevent the within-pair inference used by the figures.

C.2 Paired tests and plotting units

For modern estimator M , define $\hat{\Delta}_M = \hat{\theta}_M - \hat{\theta}_T$. Its standard error uses the shared-draw covariance:

$$\widehat{\text{SE}}(\hat{\Delta}_M) = [\widehat{V}_M + \widehat{V}_T - 2\widehat{C}_{MT}]^{1/2}. \quad (6)$$

A filled marker requires $|\hat{\Delta}_M / \widehat{\text{SE}}(\hat{\Delta}_M)| > 1.96$ and no inference-warning flag. Plotted points are

$$x = \frac{\hat{\theta}_T}{\widehat{\text{SE}}(\hat{\theta}_T)}, \quad y = \frac{\hat{\theta}_M}{\widehat{\text{SE}}(\hat{\theta}_T)}. \quad (7)$$

The denominator is the saved bootstrap standard error for that pair's baseline or aligned TWFE coordinate in outcome-SD units; its bootstrap calculation includes variation in the outcome scale. The vertical coordinate is not the modern estimator's own t statistic, and $y - x$ is not the paired-difference t statistic. Dividing both point estimates by a common positive number preserves equality and signs. This final plotting transformation does not change the underlying inference.

Each coordinate is clipped separately at -5 or 5 for display. Capped points use diamonds, with an outer diamond surrounding capped event-study squares; a diamond's apparent distance from the diagonal need not equal its actual gap, and points capped at the same corner can differ substantially. The uncapped estimates, rather than plotted coordinates, determine signs and tests. Colors identify anonymous studies, with publication years retained and four chronological legend columns. Anonymous study IDs are assigned from the 24-study source release and are not renumbered after screening. Adding earlier-year studies can renumber later studies between releases; the current figures use the updated chronological IDs, with 6, 6, 5, and 5 retained studies in the four legend columns.

D Concentration screen and descriptive-count definitions

The screen measures how an estimator’s design weights are concentrated across its published dependence clusters. For each actual scalar or pooled contrast, express the raw-outcome statistic as $\hat{\theta} = \sum_i a_i Y_i$. Contributions from repeated appearances of the same original observation are first netted at that observation before squaring. Define

$$L_c = \frac{\sum_{i \in c} a_i^2}{\sum_i a_i^2}, \quad G_{\text{eff}} = \frac{1}{\sum_c L_c^2}, \quad L_{\text{max}} = \max_c L_c. \quad (8)$$

The common outcome scale cancels from these shares.

An estimator pair is excluded if either plotted estimator has $G_{\text{eff}} < 5$ or $L_{\text{max}} > 0.5$. A retained pair receives an inference warning if either estimator, or their nonidentity difference, has $G_{\text{eff}} < 10$ or $L_{\text{max}} > 0.25$. Earlier inference warnings are also retained. Warning-marked points are crosses.

Table 3: Counts underlying the figure annotations

	Baseline		Aligned	
	CSA	BJS	CSA	BJS
Available before screen	369	373	369	373
Excluded by screen	96	76	96	76
Retained pairs	273	297	273	297
Included pre-treatment averages	22	22	22	22
Warning-marked pairs	68	82	66	75
Opposite nonzero signs	37	23	45	19
Significant paired gap (filled)	23	42	21	30
Significant gap and opposite signs	4	3	0	3
At least one capped coordinate	32	43	31	41

For all three annotated shares, the denominator is the number of retained estimator pairs, including warning-marked pairs. “Sign-reversals” means $\hat{\theta}_T \hat{\theta}_M < 0$. “Significant differences” counts filled points, and “Significant + Reversal” is the intersection of those two categories. Warning-marked points can contribute to sign reversals but cannot contribute to either filled-point numerator. Their absence from the significant count is not evidence of a precise zero gap. No share is weighted to give each study equal influence, and no pooled independent-specification test is implied. The reported relative standard errors are arithmetic means of the modern-to-TWFE standard-error ratio over retained pairs in each panel, including pre-treatment averages and warning-marked pairs; they are not ratios of mean standard errors or comparisons of published standard errors.

E Implementation exceptions and reproduction records

The following records explain departures from a literal “published TWFE versus modern estimator on the same sample” description. They use only the anonymous study IDs shown in the exhibits.

Fixed-effect specification (Study 8). The literal source code omits an intercept and one indicator from each fixed-effect set, imposing an additional restriction at the omitted-category intersection. Its coefficients and robust standard errors reproduce the published rounding and

are preserved separately. The plotted TWFE comparator uses the full unrestricted additive fixed-effect span under the recorded implementation exception. The bootstrap follows the published observation-level robust dependence rule using 1,048 state-year rows, rather than eight state clusters.

Conflicting records and dependence partitions (Study 2). The analysis preserves the original rows and fixed-effect identifiers even where a fixed-effect-unit/time cell contains conflicting treatment or outcome records. Separately audited finer treatment histories determine adoption cohorts, including a joint firm-identifier history where needed. CSA uses these histories, while BJS retains the published fixed-effect structure. The published state and state-by-year clustering partitions remain separate covariance blocks.

Native modern estimates and constructed TWFE (Studies 18, 22, and 23). Study 18 contributes both published TWFE level specifications and published CSA log-outcome specifications. The latter receive constructed TWFE comparisons. The comparison CSA uses four-cell repeated-cross-section means with period-specific published analysis weights; its separately preserved native CSA uses a panel formulation. Dependence is clustered by county. Study 22’s native CSA variants include never-treated and not-yet-treated donors and their native baseline conventions; the comparison implementation uses its registered not-yet-treated donor convention. Source variants yielding the same normalized comparison are counted once. Study 23’s native BJS event studies receive constructed joint TWFE comparisons with the authored indicators preserved.

Event windows and mixed references (Studies 19 and 20). Study 19 contains the sole retained event comparison whose common post window changes between figure families, as detailed in Appendix B. Its strictly pre-treatment summary uses years -3 and -2 . Study 20 preserves the published mixed pre/post reference bin in the modern contrasts and their bootstrap draws. Neither implementation replaces missing horizons with zero.

References

- [1] Callaway, Brantly, and Pedro H. C. Sant’Anna. 2021. “Difference-in-Differences with Multiple Time Periods.” *Journal of Econometrics* 225(2): 200–230. doi:10.1016/j.jeconom.2020.12.001.
- [2] Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2024. “Revisiting Event-Study Designs: Robust and Efficient Estimation.” *Review of Economic Studies* 91(6): 3253–3285. doi:10.1093/restud/rdae007.